

Generation Interconnection System Impact Study Report

For Queue Project Q13-0210

Mooney 138 kV

March 22, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0210. The Interconnection Customer proposes a 106.3 MW solar photovoltaic facility interconnecting at the Mooney 138 kV substation in the Maryland area. The request was received on March 06, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 128 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,416,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HUTCHINSON PLANT 1 - WEST KELSO 138 kV L	ALTAVISTA POWER STATION - RUBIDOUX 138 kV	142.3	214	6.6
ALTAVISTA POWER STA - RUBDOUX SUBSTATION 138 kV	WEST KELSO - DRY GAP 138 kV	133.7	201	9.6
W KELSO - DRY GAP 138 kV	ALTAVISTA POWER STA - RUBIDOUX 138 kV	128.9	246	6.6
WATFORD CITY - HUTCHINSON PLANT 1 SUBSTATION 138 kV	FOUR HILLS - MORGAN FALLS 138 kV L	127.6	220	7.2
BASSETT - WALKERSVILLE 138 kV	ARTEIA COUNTRY CLUB - CLARK COUNTY 345 kV L	114.6	196	26.3
FOUR HILLS - RUBIDOUX 138 kV	W KELSO - DRY GAP SUBSTATION 138 kV L	111.8	218	9.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST KELSO - HUTCHINSON PLANT 1 138 kV	Rebuild line with higher-capacity conductor	\$9,596,000	100% (\$9,596,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALTAVISTA POWER STA - RUBDOUX SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,195,000	100% (\$2,195,000)
W KELSO - DRY GAP 138 kV	Rebuild line with higher-capacity conductor	\$3,750,000	100% (\$3,750,000)
WATFORD CITY - HUTCHINSON PLANT 1 SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$1,853,000	45% (\$835,000)
WALKERSVILLE - BASSETT 138 kV	Rebuild line with higher-capacity conductor	\$1,779,000	100% (\$1,779,000)
FOUR HILLS - RUBIDOUX 138 kV	Rebuild line with higher-capacity conductor	\$17,261,000	100% (\$17,261,000)
	Total allocated to project		\$35,416,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.